Interface Design

Lecture # 22, 23, 24

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SE1001 –Intro. To Software Engineering

TODAY'S OUTLINE

- **Interface Design**
- **Introduction**
- **Importance of interface Design**
- **Interface Design Errors**
- **Golden rules for interface design**

INTERFACE DESIGN

- Initially users must adapt to complex systems
  - For being technical
- Now technology must consider human ease
- Usability matters a lot!!!!

INTERFACE DESIGN

- Interface design is not just about the arrangement of media on a screen
- It's designing an entire experience for people, hence a "look and feel"
- **Easy to learn**
- **Easy to use**
- **Easy to understand**

IMPORTANCE OF INTERFACE DESIGN

- System users often judge a system by its interface rather than its functionality
- A poorly designed interface can cause a user to make catastrophic errors.
- Poor user interface design is the reason why so many software systems are never used

WHY IS GOOD USER INTERFACE DESIGN IMPORTANT?

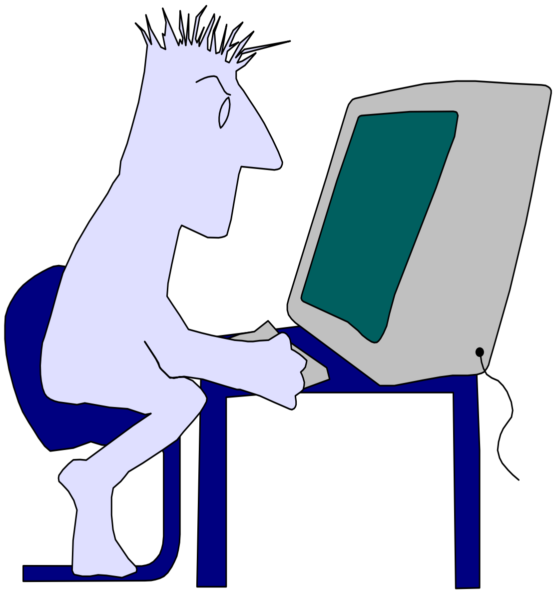
"Interface inconsistency can cost a big company millions of dollars in lost productivity and increased support costs."

Jesse Berst (1993)

- **Reduced number of users**
- **Company will face financial loss**
- **Designer's profile will be affected**

INTERFACE DESIGN ERRORS

- **lack of consistency**
- **too much memorization**
- **no guidance / help**
- **poor response**
- **unfriendly**



GOLDEN RULES

- Don't do to others what others have done to you. Remember the things you don't like in software interfaces you use. Then make sure you don't do the same things to users of interfaces you design and develop.
- The three areas of user interface design principles are:

1) **Place the user in control**
2) **Reduce the user's memory load**
3) **Make the interface consistent**

GOLDEN RULES FOR UI DESIGN (BY THEO MANDEL)

1. Place the user in control
2. Reduce user's memory load
3. Make a consistent interface

GOLDEN RULE # 1
PLACE THE USER IN CONTROL

1. INTERACTION MODES

- Define interaction modes in a way that does not force a user into unnecessary or undesired actions
- Two types of interaction modes
- The first type is **application modal**. When in an application mode, users are not allowed to work elsewhere in the program, or it places them in a certain mode for the whole program.
- For example, if working with a database of information and the **viewing mode** is chosen, the program will not allow users to add, delete, or **edit the data** record.
- It is very important that, when designing an interface, you include a visual indicator showing the user the current mode.

INTERACTION MODES

- The second type of mode is ***system modal***. *While in a* system mode, users are not allowed to work anywhere else on the computer until the mode is ended, or it places them in a certain mode no matter what program they are using.
- The user shall be able to enter and exit a mode with little or no effort
- Example: a document is **printing**, and a message window pops up stating the **printer is out of paper**. Users should not have to get up and put paper in the printer immediately, or even remove the message from the screen. They should be able to continue working. Users might even want to keep the message window on the screen as a reminder to add paper later.

2. PROVIDE FLEXIBLE INTERACTION

- The user shall be able to perform the same action via keyboard commands, mouse movement, or voice recognition

3. UNINTERRUPTIBLE USER INTERACTION

- **Don't force users to complete predefined sequences. Give them options—to cancel or to save and return to where they left off.**
- **The user shall be able to easily interrupt a sequence of actions to do something else (without losing the work that has been done so far)**
- **"Wizards" are used more and more to lead users through common tasks, but don't lead with an iron hand, let users stay in control while the interface *guides them rather than forces them through steps in a task.***

4. PROVIDE REVERSIBLE ACTIONS, AND IMMEDIATE FEEDBACK

- Provide undo actions for users, and hopefully, also redo actions. Inform users if an action cannot be undone and allow them to choose alternative actions, if possible.
- Provide users with some indication that an action has been performed, either by showing them the results of the action, or acknowledging that the action has taken place successfully.

5. PROVIDE SIMPLE NAVIGATION

- **Provide meaningful paths and exits**
- **Users should be able to relax and enjoy exploring the interface of any software product. Users should not be afraid to press a button or navigate to another screen.**
- **Navigation is a simple process to learn—basically, navigation is the main interaction technique on the Internet.**

6. STREAMLINED INTERACTIONS

- **Streamline interaction as skill levels advance and allow the interaction to be customized**
- **Don't sacrifice expert users for an easy-to-use interface for casual users. You must provide fast paths for experienced users.**
- **For example, the menu bar and pull-downs of a program can be set up as "standard" or "advanced", depending on user preferences and the types of tasks being performed.**
- **The user shall be able to use a macro mechanism to perform a sequence of repeated interactions and to customize the interface.**

7. HIDE TECHNICAL INTERNALS FROM THE CASUAL USER

- The user shall not be required to directly use operating system, file management, networking. etc., commands to perform any actions. Instead, these operations shall be hidden from the user and performed "behind the scenes" in the form of a real-world abstraction

- The interface can be made transparent by giving users work objects rather than system objects. Trash cans, waste baskets, shredders, and in- and out-baskets all let users focus on the tasks they want to do using these objects, rather than the underlying system functions actually performed by these objects.

8. DESIGN FOR DIRECT INTERACTION

- **The user shall be able to manipulate objects on the screen in a manner similar to what would occur if the objects were a physical thing (e.g., stretch a rectangle, press a button, move a slider)**
- **Users feel more comfortable and in control of the interface if they can personalize it with their favorite colors, patterns, fonts, and background graphics for their desktop.**

GOLDEN RULE # 2
REDUCE THE USER'S MEMORY LOAD

1. RELIVE SHORT TERM MEMORY

- The system should be able to retrieve the previous information so users don't have to remember and retype the information again.
- For example, when filling in online forms, customer names, addresses, and telephone numbers should be remembered by the system once a user has entered them, or once a customer record has been opened.

2. Rely on Recognition, Not Recall

- People have a limited short-term memory—we can 'instantaneously' remember about seven items of information (Miller, 1957). Therefore, if you present users with too much information at the same time, they may not be able to take it all in.
- Provide lists and menus containing selectable items instead of fields where users must type in information without support from the system.
- E.g. file menu, edit menu

ESTABLISH MEANINGFUL DEFAULT

- **The system shall provide the user with default values that make sense to the average user but allow the user to change these defaults**
- **The user shall be able to easily reset any value to its original default value**
- **E.g. Reset Option**

REAL WORLD METAPHOR

- **The visual layout of the interface should be based on a real world metaphor**
  - **The screen layout of the user interface shall contain well-understood visual cues that the user can relate to real-world actions**
  - **For example, a bill payment system should use a checkbook and check register metaphor to guide the user through the bill paying process. This enables the user to rely on well-understood cues, rather than memorizing an interaction sequence.**

DEFINE SHORTCUTS

- Define shortcuts that are intuitive
  - The user shall be provided mnemonics (i.e., control or alt combinations) that tie easily to the action in a way that is easy to remember such as the first letter.
  - E.g. Alt P for printing

PROVIDE VISUAL CUES (INFORM)

- Whenever users are in a mode, or are performing actions with the mouse, there should be some visual indication somewhere on the screen that they are in that mode.
- Test a product's visual cues—walk away from the computer in the middle of a task and come back sometime later. Look for cues in the interface that tell you what you are working with, where you are, and what you are doing.
- MS Word, MS Power Point

DISCLOSE INFORMATION IN A PROGRESSIVE FASHION

- When interacting with a task, an object or some behavior, the interface shall be organized hierarchically by moving the user progressively in a step-wise fashion from an abstract concept to a concrete action
- e.g., text format options → format dialog box

The more a user has to remember, the more error-prone interaction with the system will be

GOLDEN RULE # 3
MAKE THE INTERFACE CONSISTENT

MAINTAIN INFORMATION IN CONSISTENT FASHION

- **The interface should present and acquire information in a consistent fashion**
- **Consistency means that users should see information and objects in the same logical, visual, or physical way throughout the product**
  - All visual information shall be organized according to a design standard that is maintained throughout all screen displays
  - Input mechanisms shall be constrained to a limited set that is used consistently throughout the application
  - Mechanisms for navigating from task to task shall be consistently defined and implemented

CHANGING PAST USER EXPECTATIONS

- **If past interactive models have created user expectations, do not make changes unless there is a compelling reason to do so.**
- Once a particular interactive sequence has become a de facto standard (e.g., alt-S to save a file), the application shall continue this expectation in every part of its functionality

MAINTAIN CONSISTENCY ACROSS A FAMILY OF APPLICATIONS

- **A set of applications performing complimentary functionality shall all implement the same design rules so that consistency is maintained for all interaction**

- **Learning how to use one program should provide positive transfer when learning how to use another similar program interface.**

DRAW THE INTERFACE DESIGN OF YOUR COURSE PROJECT ACCORDING TO THE GOLDEN RULES THAT YOU HAVE STUDIED

UI ANALYSIS & DESIGN

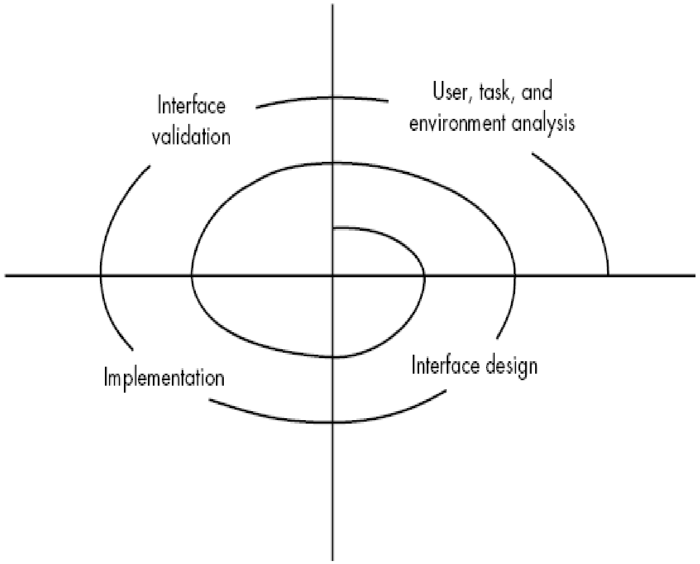
1. Segregate human-oriented and computer-oriented tasks
2. Think of design issues for UI design
3. Tools are used to prototype
4. Implement design model
5. Users evaluate the design quality

UI DESIGN MODELS

User model	Design model	System perception/mental model	Implementation model
Analyze all end user profile **Like age, gender, physical abilities, education, culture, preferences etc.** **User types:** **•Novice** **•Knowledgeable, intermittent users** **•Knowledgeable, frequent users**	**Design realization of end user's model**	**End user's perspective of a system**	**The interface look & feel along with supporting information (books, manuals, videotapes etc.) showing interface semantics.**

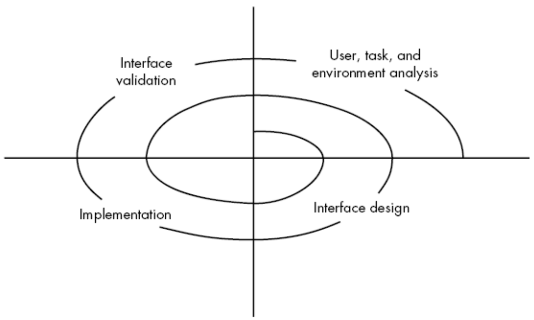
Hence, an effective design is one in which the user mental model and implementation model coincides

UI ANALYSIS & DESIGN PROCESS



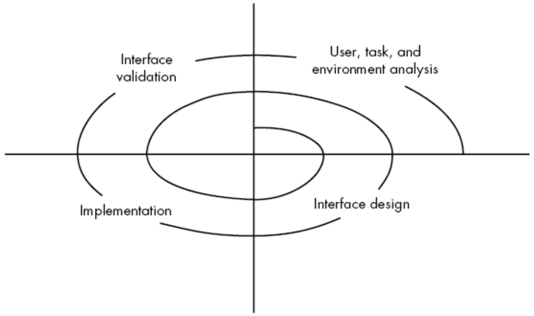
- Iterative process – It uses the spiral process model
- Four steps are:
  - Interface analysis
  - Interface design
  - Interface implementation/construction involves a prototyping approach
  - Interface validation
- Since it's spiral model so these steps need to be processed again and again for design betterment.

UI ANALYSIS & DESIGN PROCESS



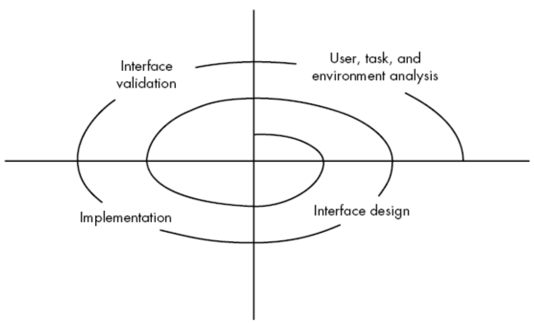
- <u>User analysis:</u>
  - who are going to use the system?
  - What is the skill level, business perception ?
  - User diversity?
    - Gather requirements for each user category & understand system by each category perspective.

UI ANALYSIS & DESIGN PROCESS



- <u>Interface design:</u>
  - A set of interface objects and actions along with their screen representation that allows user to perform all tasks and fulfil the usability goals of the system.
- <u>Interface Construction/implementation:</u>
  - Create prototype against a use case scenarios.

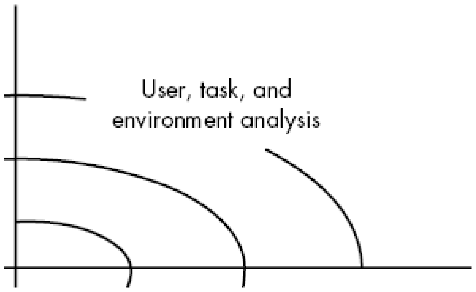
UI ANALYSIS & DESIGN PROCESS



- Interface validation:
  - Check if the system accommodates all variety of tasks?
  - Whether the system is easy to use & learn?
  - users' acceptance of a system as useful tool
- Now, we'll cover each separately in detail

INTERFACE ANALYSIS

- *User Analysis:*
- Understand problem before designing the solution.
  - Understand users of the system,
  - the system goals (tasks),
  - the displayed content,
  - the environment in which tasks is conducted.
- The mental image and the design model must converge.
  - Can be achieved by understanding the users and how they use the system



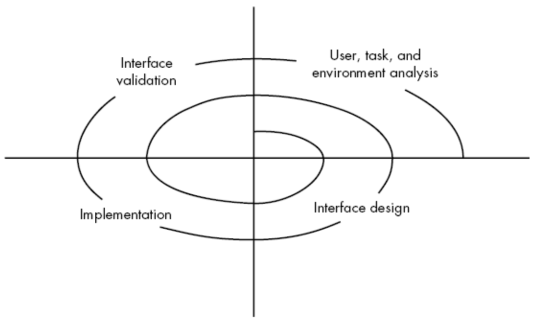
INTERFACE ANALYSIS

- Information for system understanding can be obtained by:
  - *User interviews*
    - development team members met with end users to identify their needs and work culture.
  - *Marketing & sales input*
    - Take the market survey and check in how many ways the product could come handy.
  - *Support input*
    - Support staff takes feedback of the what work or what not. Which features are easy to use or not etc.

USER ANALYSIS QUESTIONS FOR BETTER USER CATEGORIZATION

- Are **users trained professionals, technician, clerical, or manufacturing workers**?
- **Level of formal education** of an average user?
- Are the users capable of learning from written materials or have they expressed a desire for classroom training?
- users **expert typists / keyboard phobic**?
- **age range** of the user community?
- User's **gender**?
- How are users compensated for the work they perform?
- Working hours: **normal office hours / until the job is done**?
- **Software usage frequency** ? (mostly/occasionally)
- **primary spoken language among users**?
- **consequences of a user's mistake while using the system**?
- Are users experts in the subject matter that is addressed by the system?
- Are users interested in learning about the technology that sits behind the interface?

UI ANALYSIS & DESIGN PROCESS



- task analysis:
  - Detailed task analysis done after user analysis.
  - The tasks that users will perform to accomplish system goals.
- User environment analysis:
  - done to check if the current physical environment be able to cater the system needs?
- Output: Analysis model for interface

INTERFACE ANALYSIS

- *Task Analysis & Modelling*
- answer the given questions …
  - Task identification in specific circumstances.
  - Tasks and sub tasks performed by user
  - sequence of work tasks—the workflow
- Use-cases:
  - define basic interaction of actor and system
  - written using informal paragraphs.
  - Can be used to derive tasks, sub tasks and interfaces
- Task elaboration:
  - Stepwise elaboration or task refinement of interactive tasks
  - Derive either manually or use a preexisting system to identify them
- Object elaboration:
  - extract physical objects from system to define their classes and behavior
- Workflow analysis:
  - How defines how a work process is
  - a work process is completed when several people (and roles) are involved
  - Shown using UML swim line diagram

INTERFACE ANALYSIS

- *Analysis of Display Content*
  - Format of the content
  - Aesthetics
  - Content displayed using stepwise refinement approach

- *Analysis of Physical Work Environment*
  - Work environment should be helpful in proper operation and concentration .
  - Must co-relate with the designed software aesthetics.

LIBRARY MANAGEMENT SYSTEM

Task Identification

Scenario: A student wants to borrow a book from an online library system.

Main Task: Borrowing a book
Subtasks:

- User logs into the system
- searches for the desired book
- Selects the book and checks availability
- Requests book for borrowing
- Receives confirmation

Use-Case: Borrow a Book
Actor: Student
Pre-condition: The student must be logged in

Steps:

1. The student searches for a book.
2. The system displays book availability.
3. The student selects the book and requests to borrow it.
4. The system confirms the request and updates the book status.

- **Post-condition:** The book is marked as "borrowed" under the student's account.

Task Elaboration (Stepwise Breakdown of Interactive Tasks)

Stepwise Refinement of "Search for a Book" Task:

1. Open the library website.
2. Navigate to the search bar.
3. Enter the book title or author's name.
4. Click the search button.
5. System retrieves and displays results.

- This breakdown ensures a structured approach to UI and user actions.

LIBRARY MANAGEMENT SYSTEM

Object Elaboration (Extracting Physical Objects for Class Design)

Objects in the Library System:

- **Book** (Title, Author, ISBN, Availability)
- **User** (Name, ID, Borrowed Books)
- **Librarian** (Manages book inventory)

📌 **Example of Object Definition in UML:**

Class: Book

Attributes: title, author, ISBN, availability

- Methods: checkAvailability(), borrowBook(), returnBook()

Workflow Analysis (How the Work Process is Completed)

Borrow Book Process – Swimlane Diagram (Roles & Actions)

- **Student:** Logs in ⟶ Searches book ⟶ Requests borrow
- **Library System:** Displays availability ⟶ Confirms borrow request
- **Librarian:** Approves request (if required) ⟶ Updates inventory

Analysis of Display Content

Format:

- Search bar, filter options for book categories
- Clear "Borrow" and "Return" buttons
- Due date notification displayed prominently

Aesthetics:

- Simple UI with readable fonts
- Step-by-step borrowing process displayed

Stepwise Refinement:

- After login → Homepage shows search bar
- After selecting a book → Book details page appears
- After borrowing → Confirmation message appears

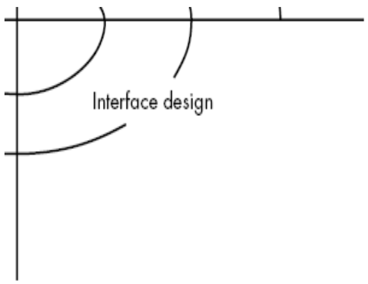
Analysis of Physical Work Environment

Considerations:

- **User-Friendly Design:** Should be accessible on desktop and mobile
- **Quiet Environment:** Digital transactions reduce noise in the library
- **Self-Service Kiosks:** Physical integration with the online system for quick book check-in/check-out

INTERFACE DESIGN STEPS

- UI Design models state that:
  - Using information developed during interface analysis,
    - define interface objects (buttons, text fields, video etc.) and operations (tasks)
    - Define events (user actions) that will change the state of the UI.
    - Depict each interface state as it will actually look to the end-user. (using sketches).
    - Show how the user infer the System state from information provided through the interface.



INTERFACE DESIGN STEPS

- Apply interface design steps
  - Define the use-case:
    - Create the list of objects and actions
    - These objects & actions are categorized into different type.
    - Identify which are source (e.g., printer icon), target (e.g., paper) and application objects(e.g., screen, entity/db table).
- Use UI design patterns
  - As specific deign pattern is formulated for a well bounded solution.
    - E.g., **for a calendar UI**, a Calendar Strip is created that allows to traverse in future dates, marking the current one.

INTERFACE DESIGN STEPS

- Design Issues
  - Some of the issues in interactive systems are:
    - System Response time
    - Help facilities
    - Error handling
    - Menu & command labelling

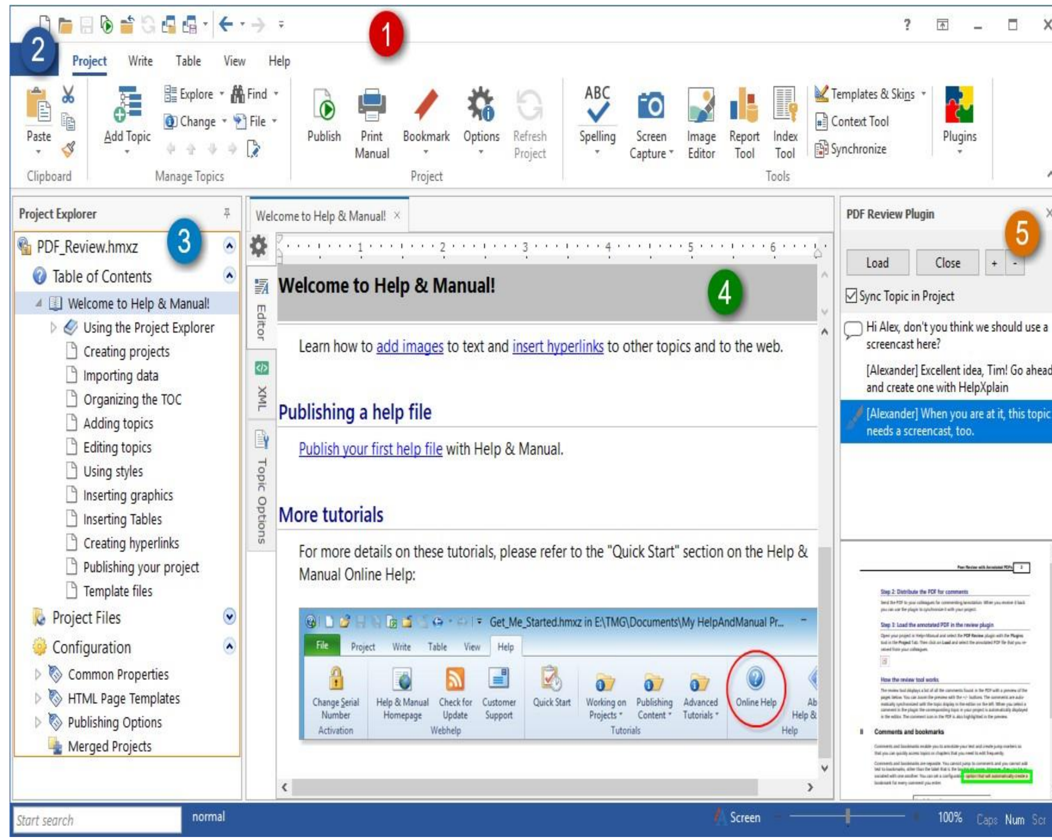
INTERFACE DESIGN STEPS

- Design Issue 1: System Response time
  - Primary complaint for almost every interactive system
  - Two characteristics:
    - Length: longer response times are not appreciated, led to user frustration
    - Variability: varying response time led to user confusion, like whether something went wrong at backend which is leading to a different response time than average response time.

INTERFACE DESIGN STEPS

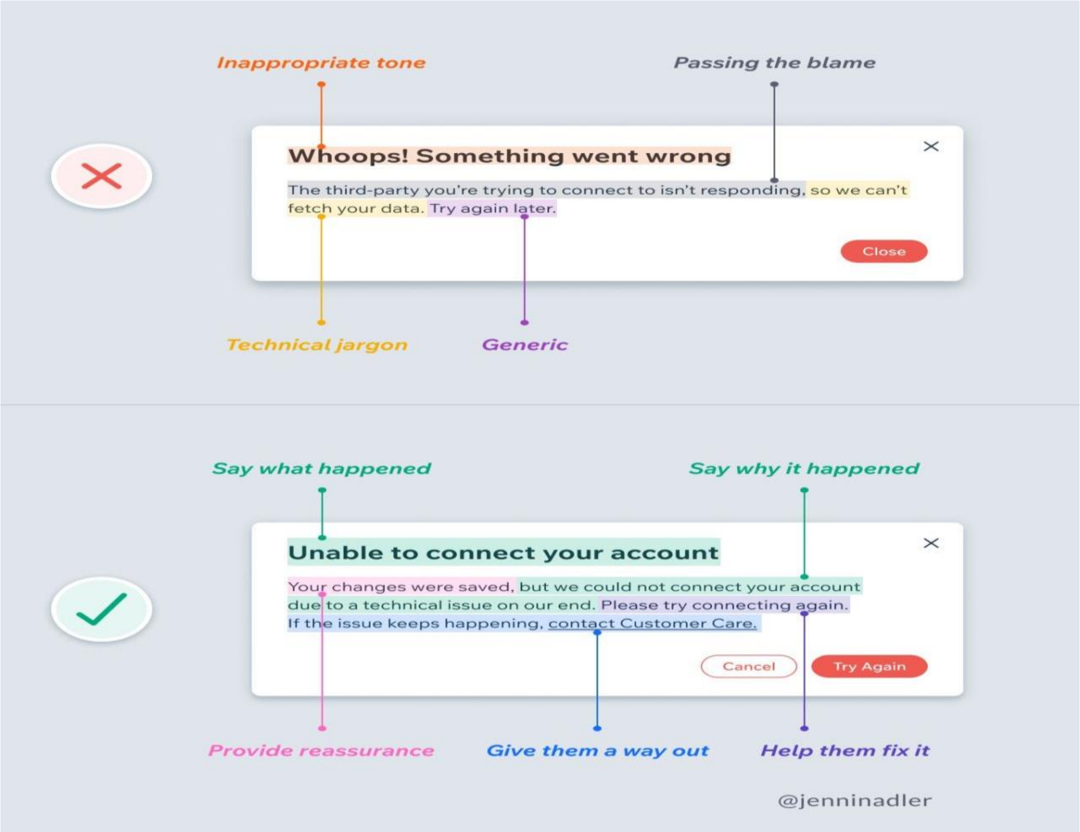
- Design Issue 2: Help Facilitates
  - Online helpdesk
  - Detailed user manuals
  - Design issues while providing Help facilities
    - **Help availability** for all functionalities?
    - **How will user demand help**?

      (give menu for that or a function key mapped to that)
    - **How is help presented**? (via another window or a printed manual)
    - **How to return to normal UI after consulting help**



INTERFACE DESIGN STEPS

- Design Issue 3: Error Handling
  - Error messages & warnings are bad news to users when something went wrong.
  - Error message must be **explanatory** and should **include some remedies** for that (via hyperlinks)
  - A **good error message** includes:
    - Describe problem simply
    - Provide valuable advice
    - List the negative consequences of the error
    - Message must prompt visually using a color scheme that easily categorize it as an error.



INTERFACE DESIGN STEPS

- Design Issue 4: Menu & Command labeling
  - Different styles like command oriented arch and UI based arch.
  - Both are used frequently
  - How commands are assigned to each menu item so that it may run without UI as well.
- Design Issue 5: Application Accessibility
  - UI should cater the needs of the physically challenged persons as well (<u>like some sort of voice inputs could be accepted to facilitate handicapped</u>)
- Design Issue 6: Internationalization
  - UI must target the global market
  - Follow global design standards & <u>facilitate multiple languages.</u>

WEBAPP INTERFACE DESIGN

- Every user interface—whether it is designed for a WebApp, a traditional software application, a consumer product, or an industrial device—should exhibit the usability characteristics.
- A WebApp interface design should answers three primary questions for the end user:
- Where am I? The interface should
  - (1) provide an indication of the WebApp that has been accessed and
  - (2) inform the user of her location in the content hierarchy.
- What can I do now? The interface should always help the user understand his current options—what functions are available, what links are live, what content is relevant?
- Where have I been, where am I going? The interface must facilitate navigation.

INTERFACE DESIGN PRINCIPLES AND GUIDELINES

- Effective interfaces are visually apparent and forgiving, instilling in their users a sense of control. Users quickly see the breadth of their options, grasp how to achieve their goals, and do their work.
- Effective interfaces do not concern the user with the inner workings of the system. Work is carefully and continuously saved, with full option for the user to undo any activity at any time.
- Effective applications and services perform a maximum of work, while requiring a minimum of information from users.

REVISED INTERFACE DESIGN GUIDELINES

Anticipation	Communication	Consistency	Controlled autonomy	Efficiency
Application must depicts the user's expected move. E.g., **software installation procedures are defined in a way to predict next step for users**. They don't need to search for the next work.	**Application must show the status of an activity initiated by user.** E.g., **file copying shown via progress bar**	**The navigation controls, menu, icons, aesthetics should be consistent throughout the application** E.g., like yellow triangle shows warning signs use them for warning messages only.	**Revising the golden rule that says place user in control but restrict the controls as per the user's role.** Enforce id and password for no go options.	**Design of the application must work on user's efficiency.** e.g., **input the CNIC or phone no without space** or hyphen will lead to less processing time at user end.

REVISED INTERFACE DESIGN GUIDELINES

Focus

The interface should help user stay focused and give the relevant options and data first

Fitt's Law

Law—"**The time to acquire a target is a function of the distance to the target and the size of the target.**"

Latency Reduction

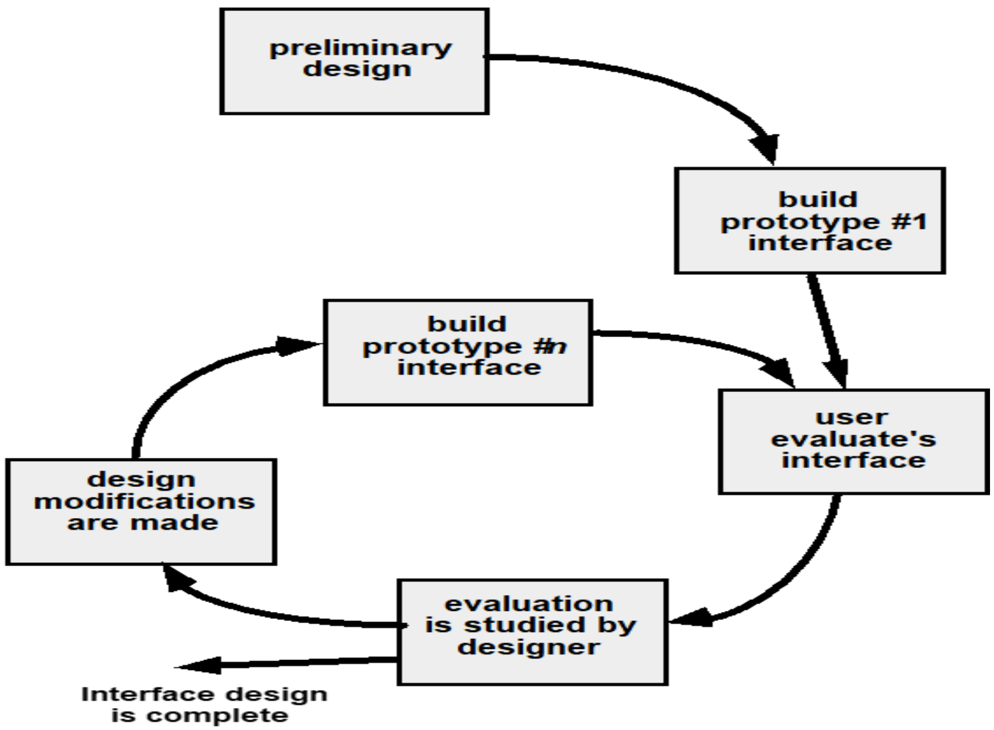
Interface must do multitasking rather than kept user waiting for an operation to complete

e.g., **display a progress bar to show the progress of the task like downloading or** Show and animated visuals to keep user focused.

INTERFACE DESIGN WORKFLOW:

- Derive and refine information from requirements
- Develop rough layout sketch
- Map user objective to actions (like Home, Contact, booking, facililities, etc….)
- Create storyboard screen images for every interface
- Refine layouts by improving aesthetics
- Develop an activity diagram to show the design flow
- Develop state diagrams to show changes in state after transitions
- Describe interface layout for every state
- Refine & review the model.

INTERFACE DESIGN EVOLUTION



That is all